

B. Third-party, prior, and acquisition conditions

Indicator scores by workflow. Each task row has CANN and CUDA columns; G uses ROCm/HIP in the second column.

| Task                                           |                            | M5 3P count |      | M6 3P credibility |      | M7 Prior estimate* |      | M8 Search / access effort |      |
|------------------------------------------------|----------------------------|-------------|------|-------------------|------|--------------------|------|---------------------------|------|
|                                                |                            | CANN        | CUDA | CANN              | CUDA | CANN               | CUDA | CANN                      | CUDA |
| Environment and installation                   |                            |             |      |                   |      |                    |      |                           |      |
| I                                              | Version compatibility      | 3           | 2    | 4                 | 4    | 2                  | 4    | 3                         | 3    |
| J                                              | Installation               | 3           | 3    | 4                 | 4    | 4                  | 5    | 5                         | 5    |
| K                                              | Container setup            | 2           | 3    | 2                 | 4    | 3                  | 5    | 1                         | 4    |
| Operator development                           |                            |             |      |                   |      |                    |      |                           |      |
| C                                              | Library selection          | 3           | 5    | 4                 | 4    | 3                  | 5    | 5                         | 5    |
| D                                              | Custom operator            | 3           | 4    | 2                 | 4    | 2                  | 5    | 1                         | 5    |
| L                                              | Operator accuracy          | 3           | 2    | 4                 | 4    | 3                  | 4    | 4                         | 1    |
| M                                              | Dynamic shapes / tiling    | 2           | 2    | 4                 | 4    | 2                  | 5    | 3                         | 5    |
| N                                              | Operator fusion            | 2           | 2    | 3                 | 4    | 2                  | 4    | 5                         | 1    |
| O                                              | Framework integration      | 3           | 2    | 5                 | 4    | 3                  | 5    | 1                         | 3    |
| Training                                       |                            |             |      |                   |      |                    |      |                           |      |
| F                                              | Distributed training       | 3           | 3    | 3                 | 4    | 3                  | 4    | 5                         | 5    |
| P                                              | Mixed precision            | 3           | 3    | 4                 | 4    | 4                  | 5    | 2                         | 4    |
| Q                                              | Memory / OOM               | 3           | 3    | 4                 | 4    | 3                  | 5    | 3                         | 2    |
| R                                              | Training convergence       | 3           | 3    | 4                 | 3    | 3                  | 5    | 3                         | 4    |
| Inference and deployment                       |                            |             |      |                   |      |                    |      |                           |      |
| A                                              | Model conversion           | 4           | 4    | 4                 | 4    | 3                  | 4    | 4                         | 5    |
| H                                              | Quantization               | 4           | 3    | 3                 | 4    | 3                  | 4    | 4                         | 5    |
| S                                              | Inference serving          | 2           | 3    | 4                 | 4    | 2                  | 4    | 4                         | 5    |
| T                                              | Dynamic inference          | 3           | 2    | 5                 | 4    | 3                  | 5    | 3                         | 1    |
| Performance and optimization                   |                            |             |      |                   |      |                    |      |                           |      |
| B                                              | Profiling                  | 3           | 4    | 3                 | 4    | 3                  | 4    | 4                         | 5    |
| U                                              | Memory / occupancy         | 3           | 3    | 4                 | 4    | 3                  | 5    | 4                         | 5    |
| V                                              | Compute-transfer overlap   | 3           | 3    | 4                 | 4    | 3                  | 5    | 4                         | 4    |
| W                                              | Multi-device communication | 3           | 3    | 4                 | 4    | 3                  | 4    | 3                         | 5    |
| Debugging                                      |                            |             |      |                   |      |                    |      |                           |      |
| E                                              | Error-code diagnosis       | 3           | 4    | 3                 | 4    | 2                  | 4    | 4                         | 5    |
| X                                              | Memory-error diagnosis     | 2           | 3    | 4                 | 4    | 2                  | 4    | 4                         | 5    |
| Migration                                      |                            |             |      |                   |      |                    |      |                           |      |
| Y                                              | Chip migration             | 3           | 3    | 4                 | 4    | 3                  | 5    | 1                         | 4    |
| Z                                              | Programming concepts       | 3           | 3    | 4                 | 4    | 4                  | 5    | 1                         | 1    |
| Migration analogy (not in CANN/CUDA summaries) |                            |             |      |                   |      |                    |      |                           |      |
| G                                              | Migration analogy          | 3           | 3    | 4                 | 4    | 3                  | 4    | 5                         | 5    |

M1–M10 are ordinal scores (1–5; higher is more favorable; M8 is an inverse effort score). “—” = unobserved official content detail.

\* M7 estimates model prior knowledge. † M11 is composite confidence from M1–M8. For G, CUDA = ROCm/HIP; it is excluded from CANN/CUDA summaries.